

Snowflake Warehouses — Developer Cheatsheet

Virtual warehouses are Snowflake's compute layer: required for queries, DML, and data load/unload. Per-second billing with a 60-second minimum each time a warehouse starts.

[!IMPORTANT] Community cheatsheet — not official Snowflake documentation. For the authoritative reference, see [Warehouses](#).

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Sizes & Credits

Gen1 credits per hour (doubles at each size step):

Size	Credits/ hour	Credits/ second	Notes
X-Small	1	0.0003	Default for CREATE WAREHOUSE (SQL)
Small	2	0.0006	
Medium	4	0.0011	
Large	8	0.0022	
X-Large	16	0.0044	

Size	Credits/ hour	Credits/ second	Notes
			Default size in Snowsight
2X- Large	32	0.0089	
3X- Large	64	0.0178	
4X- Large	128	0.0356	
5X- Large	256	0.0711	AWS + Azure GA; US Gov preview
6X- Large	512	0.1422	AWS + Azure GA; US Gov preview

[!NOTE] **Gen2** warehouses are becoming the default (BCR 2026_03, pending). Gen2 uses faster hardware and delivers better analytics and DML performance. Gen1 credit rates above apply; Gen2 rates differ — see the [Snowflake Service Consumption Table](#). X5LARGE and X6LARGE do not support Gen2. **Billing:** 60-second minimum per start. Resize while running adds new compute resources immediately but charges from the resize time.

Warehouse Types

Type	WAREHOUSE_TYPE	When to use
Standard (Gen1)	STANDARD (default)	General SQL queries, DML, data loading
Standard (Gen2)	STANDARD + GENERATION = '2'	Analytics and DML workloads; better perf than Gen1
Snowpark-optimized	SNOWPARK-OPTIMIZED	ML training, stored procs with large memory needs
Adaptive	ADAPTIVE	Analytics/ETL with variable query sizes; no sizing or scaling management needed (Enterprise, AWS preview regions)
Multi-cluster	STANDARD + cluster settings	

Type	WAREHOUSE_TYPE	When to use
		High concurrency, many parallel users (Enterprise Edition)

Create: Standard Warehouse

Minimal — defaults to X-Small, Gen1 (or Gen2 if your org/region has BCR 2026_03 enabled):

```
CREATE WAREHOUSE my_wh;
```

Common options in one block:

```
CREATE OR REPLACE WAREHOUSE my_wh
  WAREHOUSE_SIZE      = 'MEDIUM'
  AUTO_SUSPEND        = 300          -- seconds of inactivity
                                before suspending
  AUTO_RESUME          = TRUE
  INITIALLY_SUSPENDED = TRUE
  COMMENT              = 'General analytics warehouse';
```

Explicitly create a Gen2 warehouse (recommended for new warehouses):

```
CREATE OR REPLACE WAREHOUSE my_gen2_wh
  WAREHOUSE_SIZE = 'LARGE'
  GENERATION     = '2';
```

Create with query timeout and queue timeout (prevents runaway queries and queue build-up):

```
CREATE OR REPLACE WAREHOUSE my_wh
  WAREHOUSE_SIZE      = 'MEDIUM'
  AUTO_SUSPEND        = 300
  STATEMENT_TIMEOUT_IN_SECONDS = 3600  -- 1 hour max per
                                query
  STATEMENT_QUEUED_TIMEOUT_IN_SECONDS = 600  -- 10 min max in
                                queue
  INITIALLY_SUSPENDED = TRUE;
```

Key CREATE WAREHOUSE parameters:

Parameter	Default	What it controls
WAREHOUSE_SIZE	X-Small	Compute size; doubles credits at each step
WAREHOUSE_TYPE	STANDARD	STANDARD or SNOWPARK-OPTIMIZED
GENERATION		

Parameter	Default	What it controls
	Org/region-dependent	'1' or '2'; Gen2 = better analytics perf
AUTO_SUSPEND	Enabled	Seconds idle before auto-suspend; 0/NULL = never
AUTO_RESUME	TRUE	Resume on first query
INITIALLY_SUSPENDED	TRUE	Start suspended after CREATE
MAX_CLUSTER_COUNT	1	Enterprise: max clusters for scaling out
MIN_CLUSTER_COUNT	1	Enterprise: min clusters always running

Create: Snowpark-Optimized Warehouse

Use for Snowpark Python stored procedures, UDFs, and ML training workloads with large memory needs. Takes longer to start than a standard warehouse.

```
CREATE OR REPLACE WAREHOUSE my_ml_wh
  WAREHOUSE_TYPE = 'SNOWPARK-OPTIMIZED'
  WAREHOUSE_SIZE = 'MEDIUM';           -- default
  RESOURCE_CONSTRAINT = MEMORY_16X (256 GB)
```

With explicit memory/CPU configuration:

```
CREATE OR REPLACE WAREHOUSE my_ml_wh
  WAREHOUSE_TYPE      = 'SNOWPARK-OPTIMIZED'
  WAREHOUSE_SIZE      = 'LARGE'
  RESOURCE_CONSTRAINT = 'MEMORY_16X_X86'; -- 256 GB, x86
  architecture
```

RESOURCE_CONSTRAINT options for Snowpark-optimized:

Value	Memory	Min size	Availability
MEMORY_1X / MEMORY_1X_X86	16 GB	X-Small	GA, all regions
MEMORY_16X / MEMORY_16X_X86	256 GB	Medium	GA, all regions (default)
MEMORY_64X / MEMORY_64X_X86	1 TB	Large	Preview, AWS only

Create: Adaptive Warehouse

[!NOTE] **Preview feature — Enterprise Edition required.**
Available in AWS regions: US West 2 (Oregon), EU West 1 (Ireland), AP Northeast 1 (Tokyo).

With adaptive warehouses you do **not** set size, multi-cluster counts, QAS, or auto-suspend. Snowflake manages all compute automatically. All adaptive warehouses in an account share a dedicated compute pool and use per-query billing.

Create a new adaptive warehouse (minimal):

```
CREATE WAREHOUSE my_adaptive_wh
  WAREHOUSE_TYPE = ADAPTIVE;
```

With performance and throughput tuning:

```
CREATE WAREHOUSE my_adaptive_wh
  WAREHOUSE_TYPE           = ADAPTIVE
  MAX_QUERY_PERFORMANCE_LEVEL = XLARGE
  -- upper bound per query (default XLARGE)
  QUERY_THROUGHPUT_MULTIPLIER = 2      -- concurrency scale
  -- factor (default 2, 0 = unlimited)
  COMMENT                   = 'Analytics warehouse – adaptive
  compute';
```

Convert an existing standard warehouse to adaptive (no downtime):

```
ALTER WAREHOUSE my_existing_wh SET WAREHOUSE_TYPE = ADAPTIVE;
```

Key parameters:

Parameter	Default	What it controls
MAX_QUERY_PERFORMANCE_LEVEL	XLARGE	Per-query resource upper bound (XSMALL-X4LARGE)
QUERY_THROUGHPUT_MULTIPLIER	2	Concurrency scale; 0 = unlimited burst

Enable/disable accepting new jobs (does not drop running queries):

```
ALTER WAREHOUSE my_adaptive_wh SUSPEND;  -- stop accepting new
jobs
ALTER WAREHOUSE my_adaptive_wh RESUME;    -- resume accepting jobs
```

Use & Inspect

```
USE WAREHOUSE my_wh;           -- set active warehouse for
    session

SHOW WAREHOUSES;               -- list all warehouses you have
    access to

SHOW WAREHOUSES LIKE 'my_%';  -- filter by name pattern
DESCRIBE WAREHOUSE my_wh;      -- full property list

SHOW PARAMETERS FOR WAREHOUSE my_wh; -- session and warehouse
    parameters
```

Manage: Resize & Tune

```
ALTER WAREHOUSE my_wh RESUME;  -- manually start a suspended
    warehouse

ALTER WAREHOUSE my_wh SUSPEND; -- manually stop a running
    warehouse
```

Resize (takes effect for new queries; existing queries keep their resources):

```
ALTER WAREHOUSE my_wh SET
    WAREHOUSE_SIZE = 'LARGE'
    WAIT_FOR_COMPLETION = TRUE; -- wait for existing queries to
    finish before resize
```

Adjust auto-suspend and timeouts:

```
ALTER WAREHOUSE my_wh SET
    AUTO_SUSPEND = 120 -- suspend after 2
    min of idle
    STATEMENT_TIMEOUT_IN_SECONDS = 1800 -- 30 min max per
    query
    STATEMENT_QUEUED_TIMEOUT_IN_SECONDS = 300; -- 5 min max in
    queue
```

Convert an existing Gen1 warehouse to Gen2:

```
ALTER WAREHOUSE my_wh SET GENERATION = '2'; -- works running or
    suspended
```

Convert standard to Snowpark-optimized:

```
ALTER WAREHOUSE my_wh SET
    WAREHOUSE_TYPE = 'SNOWPARK-OPTIMIZED'
    RESOURCE_CONSTRAINT = 'MEMORY_1X';
```

[!NOTE] Converting a running warehouse: existing queries finish on the old type; new queries run on the new type. You are charged for both during the transition period.

Multi-cluster Warehouses

Enterprise Edition feature. Scales out by adding clusters to handle concurrency spikes automatically, without manual resizing.

```
CREATE OR REPLACE WAREHOUSE my_concurrent_wh
  WAREHOUSE_SIZE      = 'MEDIUM'
  MIN_CLUSTER_COUNT   = 1          -- minimum clusters always
                                running
  MAX_CLUSTER_COUNT    = 5          -- scale out up to 5 clusters
  SCALING_POLICY       = 'AUTO'    -- 'AUTO' (default) or
                                'ECONOMY'
  AUTO_SUSPEND        = 300;
```

Scaling policies:

Policy	Behavior	Use when
AUTO	Adds clusters to minimize queuing	Most workloads; prioritizes availability
ECONOMY	Fully loads existing clusters before adding new ones	Cost-sensitive; some query queuing is acceptable

Adjust cluster count on a running multi-cluster warehouse:

```
ALTER WAREHOUSE my_concurrent_wh SET
  MIN_CLUSTER_COUNT = 2
  MAX_CLUSTER_COUNT = 8;
```

Resource Monitors

Cap credit spend per warehouse. When the quota is hit, Snowflake can notify, suspend, or suspend immediately.

```
CREATE RESOURCE MONITOR my_daily_cap
  WITH CREDIT_QUOTA = 50          -- credits per monitoring
                                interval
  FREQUENCY          = DAILY
  START_TIMESTAMP    = IMMEDIATELY
  TRIGGERS
    ON 75 PERCENT DO NOTIFY      -- email alert at 75%
    ON 100 PERCENT DO Suspend    -- suspend new queries at 100%
    ON 110 PERCENT DO Suspend_IMMEDIATE; -- kill running queries
                                at 110%
```

Assign a resource monitor to a warehouse:

```
ALTER WAREHOUSE my_wh SET RESOURCE_MONITOR = my_daily_cap;
```

Remove a resource monitor:

```
ALTER WAREHOUSE my_wh SET RESOURCE_MONITOR = NULL;
```

Monitor Usage

Check credit consumption for all warehouses over the last 7 days:

```
SELECT
  warehouse_name,
  SUM(credits_used)           AS total_credits,
  SUM(credits_used_compute)  AS compute_credits,
  SUM(credits_used_cloud_services) AS cloud_credits
FROM SNOWFLAKE.ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY
WHERE start_time >= DATEADD('day', -7, CURRENT_TIMESTAMP())
GROUP BY 1
ORDER BY total_credits DESC;
```

Top warehouses by cost in the last 30 days:

```
SELECT
  warehouse_name,
  ROUND(SUM(credits_used), 2) AS credits_30d
FROM SNOWFLAKE.ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY
WHERE start_time >= DATEADD('day', -30, CURRENT_TIMESTAMP())
GROUP BY 1
ORDER BY credits_30d DESC
LIMIT 10;
```

Permissions

```
GRANT USAGE    ON WAREHOUSE my_wh TO ROLE analyst_role;  -- run
               queries
GRANT OPERATE  ON WAREHOUSE my_wh TO ROLE ops_role;      -- start/
               stop/resize
GRANT MODIFY   ON WAREHOUSE my_wh TO ROLE admin_role;    -- change
               parameters
GRANT MONITOR  ON WAREHOUSE my_wh TO ROLE dba_role;      -- view
               load metrics
```

Revoke:

```
REVOKE USAGE ON WAREHOUSE my_wh FROM ROLE analyst_role;
```


Drop

DROP WAREHOUSE my_wh;

[!IMPORTANT] Warehouses cannot be undropped. Verify the name before dropping.

Tips & Gotchas

- **Suspending drops the query cache.** The first queries after a warehouse resumes will be slower while the cache rebuilds. Don't suspend interactive/BI warehouses between queries if sub-second latency matters — set `AUTO_SUSPEND` to match actual gap patterns.
- **60-second minimum per start.** Stopping a warehouse in the first 60 seconds doesn't save credits — you've already been billed for that minute. Only auto-suspend after your minimum gap is consistently longer than 60 seconds.
- **Larger is not always faster.** Small/simple queries don't benefit from X-Large+. Larger warehouses help complex analytical queries, ML training, and high-concurrency loads. Test with your actual workload before sizing up.
- **Gen2 is the new default (pending).** BCR 2026_03 makes Gen2 the default for new warehouses in supported regions. New Gen2 warehouses automatically get Query Acceleration Service (QAS) enabled with `QUERY_ACCELERATION_MAX_SCALE_FACTOR = 2`. Altering a Gen1 warehouse to Gen2 does NOT auto-enable QAS.
- **Multi-cluster requires Enterprise Edition.** `MIN_CLUSTER_COUNT` and `MAX_CLUSTER_COUNT` are silently ignored on lower editions — your warehouse stays single-cluster even if you set `MAX_CLUSTER_COUNT = 5`.

References

- [Virtual Warehouses](#)
- [Overview of Warehouses](#)
- [Gen2 Standard Warehouses](#)
- [Snowpark-Optimized Warehouses](#)
- [Adaptive Compute](#)
- [Multi-cluster Warehouses](#)
- [Working with Resource Monitors](#)
- [CREATE WAREHOUSE](#)